

Two-step record creation with APIs :

an XML-based approach to creating and uploading MARC records from repository metadata for ETDs

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Background

Both authors and researchers benefit from ETDs being discoverable through basic searching on a library's discovery platform, but getting user-created metadata from a research repository into a valid MARC format can be complicated and time-consuming.

Previously, WVU Libraries utilized OCLC's Digital Collection Gateway to generate MARC from collections in the Research Repository @ WVU. However, when OCLC deprecated the product, WVU was left with poor quality records and discovered that some ETDs in the collection were never processed at all.

Approach

Many libraries have taken similar approaches to deriving MARC from ETD metadata, utilizing XSLT and batch processing. However, WVU integrates OCLC's WorldCat Metadata APIs, minimizing the amount of time that catalogers have to spend working on these records, and combining the process of retrieving, transforming, and uploading metadata into one press of a button. Our catalogers only need to update the date range and file names each semester before running the script.

Python

- Sickle
 - OAI-PMH handling
- lxml.etree
 - XML tree formatting and processing
- Saxonche
 - Saxonica python library, handles XSLT processing
- BookOps-WorldCat
 - Wrapper for OCLC's WorldCat Metadata API

XSLT

A customized script designed to process BePress' unique document-export XML metadata ontology and crosswalk it to MARC/XML following RDA standards as closely as possible.

APIs

The WorldCat Metadata APIs, available to use through a requested key from OCLC, that validate the produced MARC records and then upload valid records to OCLC, while invalid records are output into a file for further review and editing.

Results

Now, basic MARC records are discoverable through WVU's discovery platform (WorldCat). These records include titles, authors and their ORCID IDs, abstracts, subject keywords, and more, even before catalogers enhance the records.

From only title and abstract-based searching on a separate platform to a more incorporated search functionality through our main discovery system, these ETDs have become more accessible.

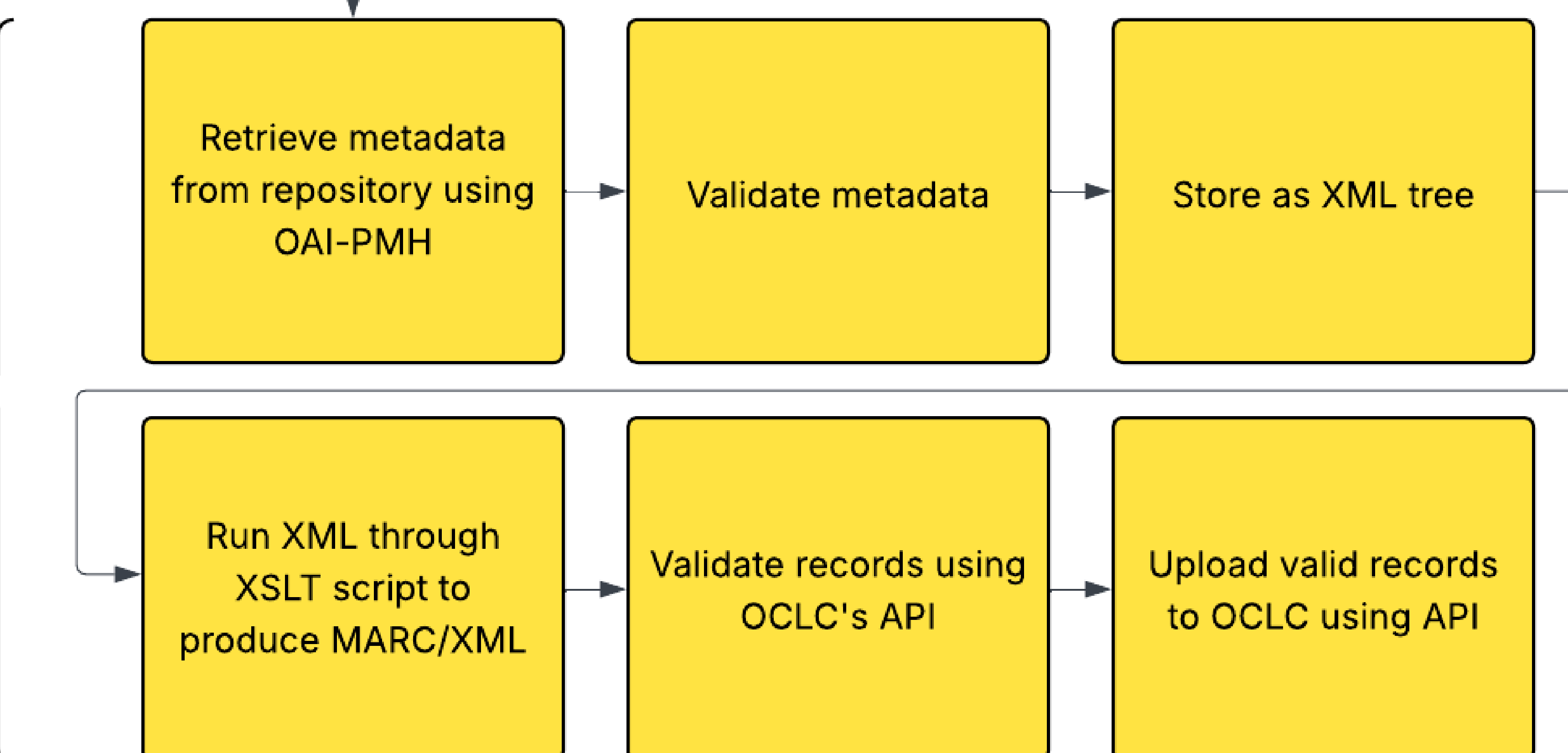
The code used to generate these MARC records is fully customizable and can be updated as standards evolve.

Next steps

There are still challenges that need to be addressed, including invalid characters present in titles and abstracts, avoiding the creation of duplicate records in OCLC, and retroactively creating records for ETDs that were missed by our previous workflow.

Additionally, these records are minimum level and only contain uncontrolled subject headings. A follow-up project is being designed that will involve catalogers working through these records to enhance and improve them.

python script



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See the code on  GitHub

Questions? Feel free to reach out!
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Do Dung Beetles Sink or Swim : Exploring [Dung Beetle](#) Response to Precipitation.

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Abstract: Global climate change constitutes a major threat to biodiversity and ecosystem services of many taxa. The Mid-Atlantic region of the US is expecting to see an increase in extreme precipitation events and flooding. [Dung beetles](#) (Scarabaeinae) can tunnel below the soil surface allowing for water to dissipate into the soil rather than sit on the surface and [Show More](#) 

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